

SecureAuth - Project Structure and Architecture Guide

This document describes the professional file structure, folder responsibilities, database design, and application flow for the SecureAuth JWT-Based User Authentication System.

Project Structure

```
SecureAuth/  
  ■■■ src/main/java/com/secureauth/  
  ■■■ config  
  ■■■ controller  
  ■■■ service  
  ■■■ repository  
  ■■■ entity  
  ■■■ dto  
  ■■■ exception  
  ■■■ resources  
  ■■■ docs  
  ■■■ pom.xml  
  ■■■ README.md  
  ■■■ .gitignore
```

controller

Receives HTTP requests such as Register, Login, Profile APIs.

service

Contains business logic like JWT generation, OTP verification, email sending.

repository

Communicates with PostgreSQL database using JPA repositories.

entity

Maps Java classes to database tables.

dto

Stores request and response objects used between client and server.

config

Contains Spring Security, JWT Filter, Password Encoder configurations.

exception

Handles application exceptions globally.

resources

Contains application.properties, templates, and static files.

docs

Stores project documents, diagrams, API documentation, and SRS files.

Database Design

Users Table

id, name, email, password, role, is_verified, created_at

Roles Table

id, role_name

OTP Table

id, email, otp, expiry_time

Application Flow

User → Register → Store User in PostgreSQL → Send Verification Email → Verify Email → Login
→ Generate JWT Token → Access Protected APIs → Manage Profile → Change Password → Logout

Recommended Technologies

Backend: Spring Boot
Security: Spring Security + JWT
Database: PostgreSQL
Email Service: Spring Mail SMTP
Build Tool: Maven
Version Control: Git & GitHub
API Testing: Postman